

A critical Besov trace obstruction to the endpoint extension in arXiv:1904.05420

Candidate counterexample packet

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Status: candidate counterexample likely valid. This is a full negative answer to the universal Besov endpoint extension in Remark 6.9, subject to human review.

1 Source problem

Caetano, Hewett, and Moiola prove in Proposition 6.7 of arXiv:1904.05420v2 [1] that, if Γ is a d -set in $\mathbb{R}^n$, $0 < d < n$, $1 < p < \infty$, $1 \leq q < \infty$, $m \in \mathbb{N}_0$, and

$$\frac{n-d}{p} + m < s < \frac{n-d}{p} + m + 1,$$

then

$$\mathcal{D}(\Gamma^c) \text{ is dense in } \ker\left(\text{Tr}_{\Gamma,m}|_{B_{p,q}^s(\mathbb{R}^n)}\right).$$

Here $\mathcal{D}(\Gamma^c) = C_c^\infty(\mathbb{R}^n \setminus \Gamma)$ and $\text{Tr}_{\Gamma,m}$ records all traces $\text{tr}_\Gamma D^\beta u$ with $|\beta| \leq m$. Remark 6.9 asks whether the same conclusion holds at the endpoint

$$s = \frac{n-d}{p} + m + 1.$$

not provide enough evidence for the statement of [32, Prop. 19.5]. Specifically, it appears to presume that $\overline{D^\beta f}(x) = 0$ μ -a.e. on Γ implies that $\overline{D^\beta f}(x) = 0$ ($s - |\beta|, p$)-q.e. on Γ for $0 \leq |\beta| \leq m$ (cf. Eqns (28) and (30) below), which a priori is not obvious to us (though it comes as a consequence for the functions f in the spaces $H_p^s(\mathbb{R}^n)$ or $B_{p,q}^s(\mathbb{R}^n)$ under the conditions of the proposition below after this has been proved).

Proposition 6.7. *Let Γ be a d -set in $\mathbb{R}^n$ with $0 < d < n$. Let $1 < p < \infty$, $1 \leq q < \infty$, $m \in \mathbb{N}_0$ and $\frac{n-d}{p} + m < s < \frac{n-d}{p} + m + 1$. Then $\mathcal{D}(\Gamma^c)$ is dense in $\ker(\text{Tr}_{\Gamma,m}|_{F_{p,2}^s})$ and in $\ker(\text{Tr}_{\Gamma,m}|_{B_{p,q}^s})$. That is,*

$$\ker(\text{Tr}_{\Gamma,m}|_{F_{p,2}^s}) = \widetilde{F}_{p,2}^s(\Gamma^c) = \widetilde{H}_p^s(\Gamma^c) \quad \text{and} \quad \ker(\text{Tr}_{\Gamma,m}|_{B_{p,q}^s}) = \widetilde{B}_{p,q}^s(\Gamma^c).$$

Proof. Since the inclusion $\supset$ is clear, we concentrate on proving the reverse one.

Step 1. $\ker(\text{Tr}_{\Gamma,m}|_{H_p^s}) \cap C^m(\mathbb{R}^n)$ approximates $\ker(\text{Tr}_{\Gamma,m}|_{H_p^s})$ in $H_p^s(\mathbb{R}^n)$.

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Proposition 6.7 in the original paper, page 29.

This finishes the proof of the proposition for the H_p^s spaces.

Step 3. Extension to $B_{p,q}^s$ spaces.

The result can be extended to the $B_{p,q}^s$ spaces by interpolation from H_p^s as in Triebel's proof in [32, pp. 262–263], using here, for each m , the s -independent (for $s \in (\frac{n-d}{p} + m, \frac{n-d}{p} + m + 1)$) bounded linear right inverses mentioned in Remark 6.4 above. $\square$

Remark 6.8. *The technique used in the above proof, of reduction to functions with enough regularity, is taken from [13, Step 1 of proof of Prop. 3.5], with a reference to [18, (i) of proof of Thm. 1]. In both [13] and [18], for the crucial part corresponding to verifying that g_j has the appropriate regularity, the reader is invited to check a related proof in [26]. By contrast, in our proof above we give a complete (and short) justification of this step using a result readily available in [17]. In the case $m = 0$ a more direct proof can also be seen in [7, proof of Prop. 2.26].*

Remark 6.9. *It is an open problem whether the result of Proposition 6.7 holds also when $s = \frac{n-d}{p} + m + 1$. If this were the case, we could update several of the results which follow by the inclusion of a corresponding limiting situation. See Remark 6.15.*

From Proposition 6.7 it is straightforward to prove the following result — which may be of independent interest — regarding a characterization of $H_{p,0}^s(\Omega)$, the closure of $C_0^\infty(\Omega)$ in $H_p^s(\Omega)$, in terms of the kernel of a trace operator.

Corollary 6.10 (Traces from $\Omega \neq \mathbb{R}^n$ to $\partial\Omega$). *Let Γ , d , n , m and s be as in Proposition 6.7, and*
The endpoint question in Remark 6.9, page 31.

2 Proof intuition

At $q = 1$, the endpoint creates one additional continuous trace. If $s = 1/p + m + 1$ for a codimension-one hyperplane, then differentiating $m + 1$ times maps $B_{p,1}^s$ into the critical space $B_{p,1}^{1/p}$, whose trace on the hyperplane is continuous. Every function supported off the hyperplane has this extra trace equal to zero. But $\text{Tr}_{\Gamma,m}$ does not record it. A bump times the monomial x_n^{m+1} therefore lies in the stated kernel while retaining a nonzero $(m + 1)$ -st normal trace. Continuity of that trace separates it from the claimed closure.

3 Counterexample

Theorem 1 (Failure of the Besov endpoint). *Let $n \geq 2$, let $\Gamma = \mathbb{R}^{n-1} \times \{0\}$, let $1 < p < \infty$, and let $m \in \mathbb{N}_0$. Set*

$$d = n - 1, \quad q = 1, \quad s = \frac{1}{p} + m + 1 = \frac{n-d}{p} + m + 1.$$

Then

$$\tilde{B}_{p,1}^s(\Gamma^c) := \overline{\mathcal{D}(\Gamma^c)}^{B_{p,1}^s(\mathbb{R}^n)} \subsetneq \ker\left(\text{Tr}_{\Gamma,m} \big|_{B_{p,1}^s(\mathbb{R}^n)}\right).$$

Consequently, the endpoint extension proposed in Remark 6.9 is false for the class of Besov spaces stated in Proposition 6.7.

Proof. The hyperplane Γ is an $(n - 1)$ -set. Choose $\eta \in C_c^\infty(\mathbb{R}^n)$ so that its restriction to Γ is not identically zero, write $x = (x', x_n)$, and define

$$u(x', x_n) = \eta(x', x_n) x_n^{m+1}.$$

Because $u \in C_c^\infty(\mathbb{R}^n)$, it belongs to $B_{p,1}^s(\mathbb{R}^n)$. If $|\beta| \leq m$, every term in $D^\beta u$ still contains a positive power of x_n . Hence

$$(D^\beta u)|_\Gamma = 0 \quad (|\beta| \leq m).$$

For smooth functions the trace operator in the source is ordinary pointwise restriction, so

$$u \in \ker \left(\text{Tr}_{\Gamma,m} |_{B_{p,1}^s(\mathbb{R}^n)} \right).$$

We show that u is not in the closure of $\mathcal{D}(\Gamma^c)$. Proposition 6.2 of the source gives a continuous critical trace

$$\text{tr}_\Gamma : B_{p,1}^{1/p}(\mathbb{R}^n) \longrightarrow L_p(\Gamma).$$

Since $D_n^{m+1} : B_{p,1}^s(\mathbb{R}^n) \rightarrow B_{p,1}^{1/p}(\mathbb{R}^n)$ is continuous, the composition

$$\Lambda(v) := \text{tr}_\Gamma(D_n^{m+1}v) : B_{p,1}^s(\mathbb{R}^n) \longrightarrow L_p(\Gamma)$$

is continuous. Every $\varphi \in \mathcal{D}(\Gamma^c)$ vanishes on a neighborhood of Γ , and therefore $\Lambda(\varphi) = 0$. It follows by continuity that Λ vanishes on $\tilde{B}_{p,1}^s(\Gamma^c)$.

Leibniz' rule, evaluated at $x_n = 0$, gives

$$\Lambda(u)(x') = D_n^{m+1}(\eta(x', x_n)x_n^{m+1})|_{x_n=0} = (m+1)!\eta(x', 0),$$

which is nonzero in $L_p(\Gamma)$ by the choice of η . Thus $u \notin \tilde{B}_{p,1}^s(\Gamma^c)$, proving strict inclusion. $\square$

The smallest concrete instance is $n = 2$, $p = 2$, $m = 0$: $\Gamma = \mathbb{R} \times \{0\}$, $s = 3/2$, and $u(x, y) = \eta(x, y)y$. Its value trace is zero, while its normal trace is $\eta(x, 0) \neq 0$.

4 Verification report

The proof uses only the following checks.

1. The chosen hyperplane satisfies every geometric hypothesis of Proposition 6.7 with $d = n - 1$.
2. The allowed parameter $q = 1$ is essential: Proposition 6.2 supplies the continuous critical trace on $B_{p,1}^{1/p}$.
3. Leibniz' rule shows all jets through order m vanish, while the pure normal jet of order $m + 1$ equals $(m + 1)!\eta|_\Gamma$.
4. Continuity of Λ makes its kernel closed and containing $\mathcal{D}(\Gamma^c)$, so it contains the claimed closure but not u .

There is no numerical or computational premise. The verifier should focus on the source's endpoint trace normalization in Proposition 6.2 and on the fact that $q = 1$ is included in Proposition 6.7.

5 Scope and literature status

This counterexample disproves the universal Besov endpoint extension; it does not contradict an endpoint theorem for Bessel-potential spaces. Hinz, Chandler-Wilde, and Hewett later cited Remark 6.9 explicitly and gave an affirmative answer for the fractional Sobolev spaces H_p^s in their Corollary 3.2 and Remark 3.3 [2]. Their paper describes the earlier Besov result but does not claim the Besov endpoint. Thus the two conclusions are compatible: the Sobolev endpoint is positive, while the Besov family in the original universal formulation already fails at $q = 1$.

6 Bounded novelty check

The search through 11 August 2026 covered the run’s registry, solution, attempt, and proof-gap indexes; exact searches for Remark 6.9 and Proposition 6.7; the title and arXiv identifier 1904.05420 combined with *endpoint*, *trace*, *Besov*, and *counterexample*; and the citing arXiv paper 2507.04536. The last source is an explicit later answer to the Sobolev branch, but the bounded search found no statement of the $q = 1$ Besov counterexample above. Novelty confidence is moderate pending a specialist search of books and non-indexed literature; mathematical validity does not depend on novelty.

7 Conclusion and review recommendation

Remark 6.9 has a negative answer as a proposed extension of all conclusions of Proposition 6.7: at the Besov endpoint with $q = 1$, the next jet becomes a continuous invariant and obstructs density. Promotion as a likely valid counterexample is recommended, with human verification of the critical trace citation and a specialist novelty check.

References

- [1] A. M. Caetano, D. P. Hewett, and A. Moiola, *Density results for Sobolev, Besov and Triebel–Lizorkin spaces on rough sets*, J. Funct. Anal. 281 (2021), 109019; arXiv:1904.05420v2. <https://arxiv.org/abs/1904.05420>.
- [2] M. Hinz, S. N. Chandler-Wilde, and D. P. Hewett, *Kernels of trace operators via fine continuity*, arXiv:2507.04536, 2025. <https://arxiv.org/abs/2507.04536>.